

MID-STATE TECHNICAL COLLEGE • IT PROGRAMS

Platform and Architecture

Unit 1 • Information Security 2 (10-151-111)



Learning Outcomes

By the end of this unit, you will be able to:



Describe the portfolio

Identify the Palo Alto Networks products that secure a network and what each one does.



Explain single-pass architecture

Describe how SP3 processes a packet once and why that matters for throughput.



Apply the Zero Trust model

Distinguish perimeter trust from Zero Trust and locate inspection points in a network.



Select a form factor

Match a physical, virtual, container, or cloud firewall to a deployment scenario.

LESSON 1

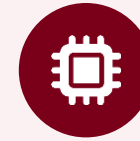
Welcome to Information Security 2



About This Course

Fifteen weeks of firewalls — mostly Palo Alto, but not only Palo Alto.

- We work primarily in **PAN-OS** on Palo Alto Networks next-generation firewalls.
- The **final project** pushes you into another vendor, so you can tell a platform from a product.
- We spend real time on **log analysis** — configuring a firewall is half the job; reading what it tells you is the other half.
- A **security research** midterm and group work round out the graded work.



How the credits split

2 credits lecture — concepts, architecture, and discussion.

1 credit lab — expect a hands-on lab nearly every week, completed on your own schedule.

Lecture builds the model; the lab makes you configure it. Skipping either one shows up fast.

Where Everything Lives

Blackboard is the source of truth for this course — not the syllabus PDF, not my slides.

- Each unit has its own **module folder**: slides, reading, lab, and the discussion prompt.
- Labs run in **NETLAB+** and are linked from the unit folder — you schedule your own pod time.
- Due dates live on the Blackboard calendar. If a date here and a date there disagree, **Blackboard is correct**.
- Slides are posted after class, so take notes on the discussion, not the bullets.

When you get stuck

Post in the unit discussion first — if one of you hit it, three others did too.

Email me for anything specific to your account, your grade, or your schedule.

LIVE DEMO

Open Blackboard and walk the Unit 1 folder, the calendar, and the NETLAB+ link together.

Why We're Here: Risk

Before we talk about firewalls at all: humans are genuinely bad at understanding risk.

- Suppose we could move people ten to twenty times faster than walking.
- Health care and emergency access improve dramatically.
- People can live somewhere other than where they work.
- Travel becomes cheap, ordinary, and even enjoyable.

The price

We must accept roughly **40,000 deaths per year** in exchange.

We already made this trade: it is the automobile. Security is not the elimination of risk — it is deciding, deliberately, which risks are worth carrying.

Every rule you write on a firewall this term is a risk decision someone will have to defend.

What a Firewall Actually Does

Strip away the marketing and the job description is short.

- A firewall **inspects** network traffic moving in and out of a network segment.
- It compares that traffic against a **policy** — an ordered set of rules you wrote.
- It then **permits or blocks** each session based on the first rule that matches.
- Everything else — threat prevention, decryption, logging — hangs off that core loop.



SECURITY FOCUS

A traditional firewall decides using port and protocol alone. Anything willing to speak on TCP/443 walks straight through.

Why it matters: A rule that trusts a port number is a rule that trusts the attacker.



LESSON 2

The Palo Alto Networks Portfolio

Four Pillars of the Approach

Palo Alto Networks organizes its entire product line around four commitments:



Complete visibility

You cannot protect what you cannot see. Visibility means users, applications, and content — across network, cloud, and endpoint.



Reduce the attack surface

Allow only what the business needs: application allow lists, acceptable-use inspection, and multi-factor authentication.



Prevent known threats

Consume threat intelligence and reconfigure defenses quickly and automatically as new signatures arrive.



Prevent unknown threats

Signatures only block what is already known. Analytics and machine learning catch behavior that has never been catalogued.

Securing the Network

The portfolio elements you will meet in this course, and what each one is for:

Element	What it is	Where it runs
NGFW	The next-generation firewall — the foundation of everything else	Physical PA-Series appliance
VM-Series	The same NGFW capabilities in a virtual machine	Private and public cloud
CN-Series	Container-native NGFW built for Kubernetes	Inside a K8s cluster
Cloud-Delivered Security Services	Subscription services that extend NGFW threat prevention	Palo Alto Networks cloud
Panorama	Centralized management, policy, and logging	Appliance or virtual

Panorama manages firewalls; it does not inspect traffic itself. Keep that distinction straight.

One Platform, Several Form Factors

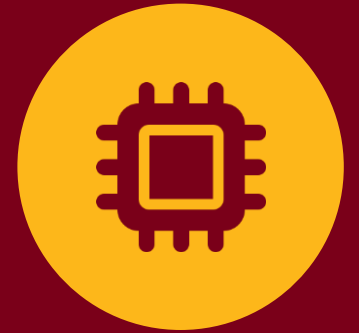
The same operating system and the same policy model, delivered three different ways.

Form factor	What it is	Typical use
Hardware PA-Series	Purpose-built physical appliances and chassis	Headquarters, data centers, branch offices
Software VM- and CN-Series	Virtual machine and container versions of the NGFW	Hybrid cloud, virtualized data center, Kubernetes
Cloud service Prisma Access	Secure access service edge delivered from the cloud	Remote workforce and distributed branches

Because PAN-OS is common across all three, what you learn on one form factor transfers to the others.

LESSON 3

Next-Generation Firewall Architecture



Beyond Ports and Protocols

What makes a firewall “next-generation” is what it uses to make the decision.

Traditional firewall

- Decides on port and protocol
- Trusts that TCP/443 is really HTTPS
- Cannot distinguish two applications sharing a port
- No concept of which user sent the traffic

Next-generation firewall

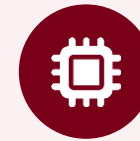
- Identifies the **application** regardless of port
- Identifies the **user**, not just the IP address
- Inspects the **content** for threats and sensitive data
- Applies all of it in a single policy

The consequence: a rule can say “allow the finance team to use this specific application” instead of “allow anything on port 443.”

Single-Pass Parallel Processing

Palo Alto Networks calls its engine SP3. The name describes two separate design choices.

- **Single pass** — networking, policy lookup, application identification, decoding, and signature matching all happen **once** per packet.
- One policy per type, evaluated in one operation, rather than a chain of separate scanning products.
- **Parallel processing** — function-specific hardware engines run at the same time rather than in sequence.
- The data plane and control plane are **separate**, so heavy management activity cannot starve traffic inspection.



Why it is fast

Traffic is scanned as it crosses the firewall with minimal buffering.

That headroom is what lets you turn on antivirus and threat prevention without watching throughput collapse.

Contrast with a chain of separate appliances, each one re-reading the same packet from scratch.

The Identification Technologies

Three identification technologies feed the policy engine on every single packet.

Technology	What it determines	How
App-ID	Which application this traffic actually is	Protocol decoding, application signatures, heuristics
User-ID	Which user or group is responsible for the traffic	Directory mapping, captive portal, multi-factor authentication
Content-ID	What is inside the traffic and whether it is safe	Threat prevention, Advanced URL Filtering, data filtering

Expect to write policy against all three by Unit 4. Learn the names now.

Control Plane and Data Plane

A Palo Alto Networks firewall is physically divided into two planes with different jobs.

Control plane

- Configuration, logging, and reporting
- Its own CPU, RAM, and **SSD** on higher-end models
- Reached through the **MGT** interface or the console
- Management load never touches traffic performance

Data plane

- Signature matching — exploits, virus, spyware, **CC#**, **SSN**
- Security processing — App-ID, User-ID, URL match, SSL/IPsec
- Network processing — flow control, route and MAC lookup, QoS, NAT
- Then it reports and enforces policy

Why the split matters: a large commit or a long report runs on the control plane, so it cannot steal cycles from the traffic being inspected.

Hardware component types and sizes per layer vary by firewall model — the architecture is what stays constant.

LESSON 4

Zero Trust Architecture



The Problem with Perimeter Trust

The conventional model assumes everything inside the network can be trusted. That assumption fails.

- Remote and mobile users are treated as internal traffic once they connect.
- Wireless clients, partner links, and guest access create ingress points the perimeter never sees.
- Remote offices may sit in places you would not otherwise trust.
- Internal employees introduce risk unintentionally — USB media, downloads, file transfers.



Where breaches start

In most breaches the attacker compromised an end-user device first, then moved laterally toward the data center. The perimeter was never the obstacle.

If you have no visibility into internal traffic, you have no control over it either.

Never Trust, Always Verify

Zero Trust removes the assumption of trust entirely — for users, devices, applications, and packets.

- No entity is trusted by default, **regardless of where it sits** relative to the corporate network.
- Trust boundaries **compartmentalize** the internal environment instead of ringing it.
- Security enforcement moves **closer to the resources** that actually need protecting.
- Verification is **continuous** — inspection keeps running after authentication succeeds.



SECURITY FOCUS

The old model was “trust but verify.” Zero Trust is “never trust, always verify” — and verify again on the next packet.

Why it matters: Authorization tells you who logged in, not what the session does an hour later.

North-South and East-West

Two traffic directions, two placements for a firewall, and historically only one of them got inspected.

North-south traffic

- Crosses the network perimeter
- Inbound from the internet, outbound to the internet
- Inspected by a firewall between your network and the internet
- The traditional, and often only, deployment

East-west traffic

- Moves between internal segments
- Users zone to applications zone, subnet to subnet, **VLAN** to VLAN
- Inspected by a firewall between internal subnets and VLANs
- Where **lateral movement** happens after a foothold

Zero Trust requires both: inspecting only what crosses the perimeter leaves an attacker free to roam once inside.

Activity: Zero Trust at the Casino

Work in your group · 15 minutes · sketch it in PowerPoint or Paint



First, draw the assets

- The vault — money
- The presidential suite — people
- The security room — data
- The floor machines — revenue-generating assets
- What else belongs on your diagram?

Then add Zero Trust controls

- Guards — people
- Obstacles, doors, and barriers — physical
- Cameras and access logs — virtual
- Where does each trust boundary sit?
- What still gets in?

A casino is a Zero Trust problem with carpet: no one inside the building is trusted by default, including staff.

Debrief: Real Risks to a Casino

Now widen the question — what could be bad enough to shut the place down?

- Consider the **physical location**, the **network**, and the **people** involved.
- Separate **internal** factors from **external** ones — insider threat looks nothing like a break-in.
- What happens during a blackout? What are your backups for electronic controls?
- What if two things go wrong at once — if you pull guards to respond, what did you just expose?

Worth raising if nobody does

Poor accounting practices causing a tax problem and a regulatory shutdown.

A ransom demand against the owner's family costing more than every cheater combined.

Remember: security is about risk. Technical controls are one instrument in a much larger score.

LESSON 5

Firewall Offerings



PA-Series Next-Generation Firewalls

A high-performance physical appliance family, from desktop units to data-center chassis.

Series	Built for	Placement
PA-7000	Power, intelligence, and versatility at scale	Enterprise and service provider
PA-5450	Hyperscale data centers and campus segmentation	Large data centers
PA-5400	No-compromise security at high performance	Data centers, service providers
PA-3400	Broad threat coverage at the internet edge	Network perimeter
PA-800	Branch offices and mid-sized businesses	Network perimeter
PA-400 / PA-410	Distributed enterprise branch requirements	Small branches, remote sites

The operating system is identical across every model, so the interface you learn in lab is the interface you meet in the field.

VM-Series Virtual Firewalls

The same next-generation firewall, delivered as a virtual machine.

- Delivers the **same capabilities** as the PA-Series hardware in a virtual form factor.
- Protects application and data traffic across **public clouds** and virtualized data centers.
- Runs on VMware, KVM, OpenStack, Cisco ACI, AWS, Azure, Google Cloud, and Oracle Cloud.
- Secures both the cloud perimeter and **east-west** traffic inside it.



Sizing changed

Older documentation sizes VM-Series by model name — VM-50, VM-300, VM-700.

Current VM-Series is sized by **allocated vCPU and memory** instead. Treat model-name specs you find online as out of date.

VM-Series Capacities

Capacity scales with the resources you allocate — representative figures:

Capacity	2 vCPU	4 vCPU	8 vCPU	16–32 vCPU
Firewall throughput (App-ID)	3 Gbps	6 Gbps	12 Gbps	24–28 Gbps
Advanced Threat Prevention	1.5 Gbps	3 Gbps	6 Gbps	12–20 Gbps
New sessions per second	15,000	30,000	60,000	120,000
Dedicated memory (minimum)	6.5 GB	9 GB	16 GB	56 GB

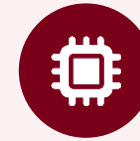
Notice that threat prevention throughput is roughly half of raw firewall throughput — inspection is not free.

CN-Series Container Firewalls

A container-native firewall that integrates directly into a Kubernetes cluster.

- Provides visibility and security for **containerized application workloads**.
- Delivers the same capabilities as PA-Series and VM-Series, in a **container** form factor.
- Offers Layer 7 visibility, application-level segmentation, DNS security, and advanced threat protection.
- Managed through **Panorama** using the Kubernetes plugin.

Protects traffic across trusted zones in public cloud and data center environments.



Why it matters

An edge firewall sees a cluster. A CN-Series firewall sees the specific pod traffic originated from.

That lets you write policy at the application level rather than the cluster level.

Virtual Systems (vsys)

Separate, logical firewall instances living inside a single physical firewall.

- Each vsys is an **independent, separately managed** firewall with its own traffic.
- Creates a genuine **administrative boundary** — useful for multiple customers or departments.
- A vsys holds its own interfaces, subinterfaces, virtual routers, and security zones.
- Supported on PA-3x00, PA-5x00, and PA-7x00; a license is required past the base count.



SECURITY FOCUS

Segmentation by vsys covers administrative access, every policy type, all objects, User-ID, certificates, and logging.

Why it matters: One box, two tenants, one shared policy — one mistake from a data disclosure.

The Security Thread in This Unit

One argument runs through every lesson.



SECURITY FOCUS

Architecture is a security control. Deciding to identify traffic by application and user rather than by port, and deciding to inspect east-west traffic rather than only the perimeter, are not performance choices — they determine what an attacker can reach after the first compromise.

Every product distinction in this unit — hardware, virtual, container, cloud — exists so that inspection can be placed where the traffic actually is.

Why it matters: A firewall in the wrong place, deciding on the wrong attribute, produces confident logs about traffic it never understood.

Carry this into Unit 2, where you will configure the box that does the inspecting.

Activity: Simulating a Firewall

Whole class · about 15 minutes · one firewall, three traffic colors, everyone else testing

- One volunteer plays the **firewall** and receives a private rule set.
- Three volunteers carry **traffic** — one color each.
- Everyone else is a **penetration tester**, directing traffic to learn which combinations pass.
- Inbound and outbound rules are not the same, which you will only discover by testing.



The point: from the outside, a policy is invisible. An attacker learns your rules the same way you just did — by probing.

Check Your Understanding

Answer these before you leave — they are the module review questions.

- Which two planes are found in the Palo Alto Networks single-pass platform architecture?
- Which object cannot be segmented using virtual systems on a firewall?
- Which series of firewalls is the high-performance physical appliance solution?
- Which product provides centralized firewall management and logging?
- True or false: CN-Series firewalls deliver the same capabilities as PA-Series and VM-Series.



If any of the five gave you trouble, review that lesson before starting the Unit 1 work.

Unit 1 Work

Always check Blackboard for the assignment list and the due dates — Blackboard wins.

- Welcome folder **survey** — tells me who you are and what you already know.
- Unit 1 **Lab: Initial Configuration** in NETLAB+.
- Security topic **discussion post** — read the SecFraudOps newsletter first.
- Palo Alto **features reading** and worksheet.



The lab covers management-interface configuration, which lines up with Unit 2 — start it once you have access confirmed.

Getting Into NETLAB+

Do this in class, before you leave the room.

```
https://netlabs.mstclab.com/
```



- Your initial **username** is your first initial and last name — no apostrophes, no spaces.
- Your initial **password** is `P@ssword01` — change it once you are in.
- You **schedule your own pod time**; reservations are first come, first served.
- Submit **one screenshot per section** of the lab unless the instructions say otherwise.

NETLAB+ is reachable from home without a VPN — no excuse to wait until you are on campus.

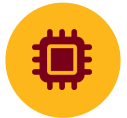
Key Takeaways



Security is risk management. Every rule you write is a risk decision.



The portfolio secures the network through NGFW, VM-Series, CN-Series, cloud services, and Panorama.



SP3 examines a packet once, in parallel, with the control and data planes kept separate.



Zero Trust means never trust, always verify — and inspect east-west, not just north-south.



One operating system, four form factors: hardware, virtual, container, and cloud service.

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Questions?

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